

INTERNSHIP PROJECT REPORT

GOOGLE PLAY STORE
ANALYTICS DASHBOARD

An End-to-End Data Analytics and Interactive Business Intelligence Project

Prepared By	Profile
Shamsiya K P	AI & Data Science Graduate
Project Domain	Technology
Data Analytics, Business Intelligence & Visualization	Python • Streamlit • Plotly

Report Purpose

This report documents the complete analytical workflow used to transform Google Play Store application data into meaningful, interactive and business-oriented insights. It covers data understanding, preprocessing, exploratory analysis, dashboard engineering, visualization, interpretation, challenges, learning outcomes and future improvements.

2026

1. Executive Summary

The Google Play Store Analytics Dashboard is an end-to-end data analytics project developed to demonstrate how raw application marketplace data can be converted into structured information, visual insights and decision-support outputs. The project combines Python-based data processing with interactive Plotly visualizations and a Streamlit dashboard interface.

The workflow begins with data acquisition and quality assessment, followed by cleaning, transformation, feature preparation, exploratory analysis and visualization. The resulting system allows users to investigate application categories, ratings, reviews, installations, pricing, content ratings and time-oriented performance patterns.

A key objective is to create a coherent analytical experience rather than a collection of unrelated charts. Interactive filters, KPI summaries, hover details, category selectors, drill-down views, normalized comparisons and time-based visualizations allow users to explore the dataset from multiple perspectives.

The dashboard contains six major analytical modules: Hexbin Density Chart, Hierarchical Sunburst Chart, Calendar Heatmap, Streamgraph, Clustered Heatmap and Radar Chart. Each visualization addresses a different analytical question and contributes to a broader understanding of application performance.

The project demonstrates practical competency in data cleaning, exploratory data analysis, feature engineering, visualization design, dashboard development and business-oriented interpretation.

Key Deliverables

- Cleaned and analytics-ready Google Play Store dataset.
- Exploratory analysis of ratings, reviews, installs, categories, pricing and application attributes.
- Six interactive visualization modules built with Plotly.
- Streamlit-based dashboard with filters, KPI summaries and interactive exploration.
- Business-oriented interpretation of marketplace patterns.
- Professional technical and internship documentation.

2. Project Background and Introduction

Mobile application marketplaces generate large volumes of structured and semi-structured information. Application metadata can reveal patterns related to user engagement, popularity, category distribution, ratings, pricing and update behavior. However, raw datasets are rarely ready for direct analysis because values can contain missing entries, duplicates, inconsistent formats and mixed units.

The Google Play Store Analytics Dashboard was developed to address this challenge through a practical analytics pipeline. Instead of treating individual charts as isolated outputs, the project organizes the analysis into an integrated dashboard where multiple visual perspectives can be connected through common analytical dimensions.

Problem Statement

The central problem is to transform heterogeneous application marketplace records into reliable, interpretable and interactive analytical information. The solution must support technical data processing as well as business-level interpretation while remaining accessible through a user-friendly dashboard.

Project Scope

- Understand the structure and quality of Google Play Store application data.
- Prepare consistent numerical and categorical variables for analysis.
- Explore relationships among ratings, reviews, installs, size, price and categories.
- Analyze category-level and time-oriented installation patterns.
- Compare free and paid application performance using normalized indicators.
- Present findings through interactive visualizations and dashboard controls.
- Document insights, limitations and development decisions.

Target Users

The dashboard can support students, data analysts, business analysts, product teams and marketplace researchers who need a structured view of application marketplace patterns. It is primarily an analytical and educational project rather than a production-grade marketplace intelligence platform.

3. Project Objectives and Requirements

Primary Objectives

- Clean and preprocess application-level data to improve analytical reliability.
- Conduct exploratory data analysis to understand distributions and relationships.
- Develop interactive visualizations for different analytical dimensions.
- Create a responsive Streamlit dashboard for practical exploration.
- Identify meaningful category, rating, installation, pricing and trend patterns.
- Translate analytical results into business-oriented observations.
- Gain practical experience with a complete analytics project lifecycle.

Functional Requirements

- Load the application dataset from the project data source.
- Handle missing, duplicate, inconsistent and incorrectly formatted records.
- Convert text-based numerical fields into analysis-ready values.
- Provide interactive category and metric selections.
- Display KPI summaries and detailed visual analytics.
- Support hover-based information for detailed chart interpretation.
- Allow navigation between multiple analytical modules.
- Maintain consistent visual design and readable dashboard layout.

Non-Functional Requirements

- Readable and professional user interface.
- Reasonable responsiveness for interactive exploration.
- Reusable Python functions for loading, cleaning and transforming data.
- Clear separation between data preparation and visualization logic.
- Transparent analytical assumptions and documented limitations.

4. Dataset Description and Data Understanding

The project uses a Google Play Store application dataset containing application metadata and performance-related attributes. The exact number of records may vary depending on the dataset version used for the internship; therefore, this report focuses on the analytical schema and processing methodology rather than claiming a fixed record count.

Major Data Attributes

Field	Analytical Purpose
App / Application Name	Application identification and duplicate analysis.
Category	Category-level comparison and segmentation.
Rating	User satisfaction indicator and relationship analysis.
Reviews	Engagement and feedback volume indicator.
Installs	Application adoption and popularity analysis.
Size	Relationship with rating and adoption metrics.
Type	Free versus paid application comparison.
Price	Monetization analysis.
Content Rating	Audience segmentation.
Genres	Secondary classification and segmentation.
Android Version	Platform compatibility analysis.
Last Updated	Temporal and update-related analysis.

Data Understanding Approach

Initial inspection focuses on data types, unique values, missingness, duplicates, range validity, categorical consistency and potential outliers. Understanding these characteristics before visualization reduces the risk of misleading aggregation or chart interpretation.

5. Data Cleaning and Preprocessing

Data preprocessing is a critical stage because marketplace datasets often contain values stored as strings, special symbols, inconsistent category names, missing ratings, duplicated applications and mixed measurement units. The preprocessing workflow was designed to create consistent analytical variables while minimizing unnecessary information loss.

Cleaning Activities

- Duplicate handling: identify repeated application records and remove or consolidate them according to analytical requirements.
- Missing-value treatment: inspect nulls by field and apply appropriate removal, imputation or preservation strategies.
- Rating cleaning: convert ratings to numeric form and validate the expected range.
- Reviews conversion: convert review counts from text to numeric values.
- Installs conversion: remove symbols such as plus signs and separators before numeric conversion.
- Size normalization: convert KB/MB-style values into a consistent numerical representation.
- Price processing: remove currency symbols and prepare values for free-versus-paid analysis.
- Category standardization: normalize text values to prevent duplicate analytical categories.
- Date processing: convert update-date fields into datetime format.
- Feature preparation: create rating bands, normalized metrics, growth indicators and analytical groups.

Quality-Control Checks

After preprocessing, the dataset should be checked again for duplicate rows, invalid ranges, unexpected categories, remaining missing values and data types. Summary statistics and sample records are reviewed before the data reaches the visualization layer.

Reproducibility

Cleaning operations should be implemented through reusable Python functions rather than manual edits. This makes the workflow easier to reproduce when the dataset is replaced, updated or extended.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the statistical structure of the dataset before building the final dashboard. EDA helps identify distributions, relationships, concentration patterns, unusual records and variables that require transformation.

Univariate Analysis

Univariate analysis examines individual variables such as rating, reviews, installs, price and size. Installation counts can be highly uneven across applications, so raw linear comparisons may hide differences between high- and low-volume applications.

Bivariate Analysis

Bivariate analysis examines relationships such as rating versus reviews, size versus rating, price versus installs and category versus installations. These comparisons help identify whether engagement or popularity is concentrated in particular segments.

Categorical Analysis

Category and genre fields provide a way to compare application supply and performance across marketplace segments. Category aggregation is useful because individual application records can be noisy while broader group patterns can reveal market structure.

Temporal Analysis

Where valid update-date information is available, monthly aggregation can be used to examine changes over time. Growth calculations and moving averages help distinguish short-term fluctuations from broader directional patterns.

EDA Questions

- Which categories contain larger numbers of applications?
- How are ratings distributed across the marketplace?
- How do reviews and installations relate to application popularity?
- Are free and paid applications distributed differently across categories?
- Which categories show stronger installation concentration?
- How do application-level metrics behave over time?
- Are there unusual observations that may affect aggregated results?

7. Dashboard Architecture and Technology Stack

The dashboard follows a layered architecture in which data preparation, analytical transformation, visualization and user interaction are connected through Streamlit. This structure supports modular development and makes individual visualization pages easier to test and maintain.

Technology Stack

Technology	Role
Python	Core programming language for data processing and dashboard logic.
Pandas	Data loading, cleaning, transformation, grouping and aggregation.
NumPy	Numerical operations and feature calculations.
Plotly	Interactive charts, hover information and analytical visualization.
Streamlit	Web dashboard interface and interactive controls.
VS Code	Development, debugging and project organization.
Power BI (if applicable)	Additional business-intelligence reporting.

Application Flow

Dataset → Data Validation → Cleaning & Transformation → Feature Engineering → Analytical Aggregation → Plotly Visualizations → Streamlit Components → Interactive Exploration → Business Interpretation

Design Principles

- Consistency: common metrics and labels across charts.
- Interactivity: filters and hover details support exploration.
- Modularity: each visualization can be developed and tested independently.
- Interpretability: titles and labels communicate the analytical question.
- Performance: aggregation reduces repeated processing during interaction.

8. Visualization Module I: Hexbin Density and Hierarchical Sunburst

8.1 Hexbin Density Chart

The Hexbin Density Chart examines the relationship between application size and rating while representing concentration or installation intensity through visual density. Hexagonal bins group nearby observations, making dense regions easier to identify than an overplotted scatter chart.

Analytical Value

- Reduces overplotting when many applications occupy similar coordinate ranges.
- Makes concentration zones easier to recognize.
- Supports comparison of rating patterns across application-size ranges.
- Can incorporate installation intensity as an additional analytical signal.

8.2 Hierarchical Sunburst Chart

The Sunburst visualization presents hierarchical relationships among dimensions such as Country, Category, App Type and Rating Band. Each ring represents a deeper level of segmentation, allowing users to move from broad groups to more specific application segments.

Analytical Value

- Provides a compact view of hierarchical category structure.
- Supports drill-down style exploration through nested segments.
- Helps compare the relative contribution of application groups.
- Combines categorical segmentation with rating-based grouping.

Interpretation

A high-density region or large sunburst segment indicates concentration in the dataset, but does not by itself establish causation or explain user behavior. Additional variables and validation are required for causal conclusions.

9. Visualization Module II: Calendar Heatmap and Forecasting

9.1 Calendar Heatmap

The Calendar Heatmap presents date- or month-oriented installation activity using a color-intensity matrix. This format allows users to identify periods of relatively higher or lower activity and compare changes across time.

9.2 Month-over-Month Growth

Month-over-month growth measures the percentage change between the current period and the previous period.

$$\text{Growth (\%)} = ((\text{Current Period} - \text{Previous Period}) / \text{Previous Period}) \times 100$$

Growth metrics require careful treatment when the previous period is zero or missing; such cases should be represented as undefined rather than forcing an artificial percentage.

9.3 Three-Month Moving Average

A three-month moving average smooths short-term fluctuations by averaging the current month and the two preceding months. It provides a descriptive trend line and a simple baseline for understanding recent direction.

9.4 Forecasting Interpretation

The moving-average component should be described as a simple analytical forecast or baseline rather than as a high-confidence predictive model. Advanced forecasting would require longer time series, validation and potentially external variables.

9.5 Interactive Controls

- Category selector for focused trend analysis.
- Actual versus smoothed values.
- Custom hover information for month, category, installations and growth.
- Time-period controls for focused exploration.

10. Visualization Module III: Streamgraph and Clustered Heatmap

10.1 Streamgraph

The Streamgraph visualizes how category-level installation activity changes over time. Each flowing band represents a category, while its thickness indicates the magnitude of the selected metric. This makes it useful for observing changing category composition.

Analytical Questions

- Which categories contribute most to installation activity?
- How does category composition change over time?
- Are there periods in which a category becomes unusually prominent?
- Which categories show relatively stable or volatile patterns?

10.2 Anomaly Detection

Anomaly indicators can be derived using deviations from historical averages, rolling statistics or threshold-based rules. Such indicators are signals requiring investigation rather than proof of a business cause.

10.3 Clustered Heatmap

The Clustered Heatmap compares categories across multiple normalized performance metrics. Normalization is necessary because installations, reviews, ratings and other variables operate on different scales.

Example Metric Dimensions

- Average rating.
- Review volume.
- Installation intensity.
- Average application size.
- Average price or paid-app proportion.
- Category-level application count.

The heatmap supports pattern recognition by grouping similar performance profiles. It should be treated as a comparative visualization rather than a definitive ranking system.

11. Visualization Module IV: Radar Chart and Comparative Analysis

11.1 Radar Chart

The Radar Chart compares free and paid applications across selected normalized metrics. Because the original variables have different units and ranges, normalization is required before plotting them on a common radial scale.

Possible Comparison Dimensions

- Average rating.
- Average reviews.
- Installation level.
- Application size.
- Price-related metric.
- Other project-specific normalized indicators.

11.2 Free versus Paid Applications

The comparison provides a descriptive view of how application type is associated with selected performance indicators. A difference should not automatically be interpreted as a causal effect of pricing because quality, brand strength, category, marketing, application age and audience may also influence performance.

11.3 Dashboard KPI Layer

KPI cards can summarize major dimensions such as total applications, average rating, aggregated installations, average reviews, free-application share and paid-application share. KPI definitions should remain consistent with the cleaned dataset and aggregation logic.

11.4 Interaction Design

- Filters should update charts consistently.
- KPI values should reflect the active analytical scope where appropriate.
- Hover labels should use readable units and meaningful names.
- Chart titles should state the metric and analytical dimension.
- Color intensity should be consistent for comparable metrics.

12. Business Insights and Analytical Interpretation

The dashboard is designed to support evidence-based interpretation of marketplace patterns. The observations below describe analytical directions that can be examined using the dashboard and should be interpreted in the context of the specific dataset version used.

12.1 Category Patterns

Application categories can differ substantially in application count, installation concentration, review volume and rating distribution. Category-level aggregation therefore provides a useful market segmentation layer.

12.2 Ratings and Engagement

Ratings and reviews represent different aspects of user feedback. A rating summarizes perceived satisfaction, while review volume indicates the amount of recorded user feedback. These variables can be analyzed together to identify different engagement profiles.

12.3 Installation Concentration

Installation metrics can be concentrated among a subset of applications. Averages alone may therefore hide substantial variation. Density plots, grouped summaries and normalized heatmaps provide complementary perspectives.

12.4 Free and Paid Segments

The dashboard enables descriptive comparison between free and paid applications. Pricing type is treated as one analytical dimension among several rather than as an isolated causal explanation.

12.5 Market Opportunity Analysis

Category-level gaps between application supply, ratings, reviews and installations can serve as starting points for deeper investigation. Any opportunity hypothesis should be validated with additional variables such as competition, acquisition cost, retention and monetization.

12.6 Decision-Support Perspective

The principal value of the dashboard is its ability to move from raw records to structured comparisons. Users can identify patterns visually, inspect supporting metrics and formulate follow-up analytical questions.

13. Challenges, Solutions and Project Limitations

13.1 Major Challenges

- Inconsistent formats in installs, prices, sizes and other numeric fields.
- Missing values in important analytical variables.
- Duplicate application records that can distort aggregation.
- Categorical inconsistencies requiring standardization.
- Different metric scales complicating direct comparison.
- Designing advanced interactive visualizations without reducing readability.
- Maintaining dashboard responsiveness during filtering and transformation.
- Debugging file paths, dataset loading, chart rendering and Streamlit page integration.

13.2 Solutions Applied

- Reusable cleaning and conversion functions.
- Explicit numeric coercion and validation checks.
- Aggregation before visualization where appropriate.
- Consistent category labels and derived analytical fields.
- Normalization for multi-metric comparative charts.
- Modular Streamlit page structure.
- Interactive Plotly components with custom hover information.
- Separation of data preparation from presentation logic.

13.3 Limitations

- The dataset represents a particular snapshot and may not reflect current Google Play Store conditions.
- Observational data does not establish causal relationships.
- Simple moving-average forecasting is a baseline rather than a full predictive system.
- Some fields may contain collection or reporting bias.
- Business conclusions should be validated with current, domain-specific data before real-world decisions.

14. Learning Outcomes, Future Enhancements and Conclusion

14.1 Learning Outcomes

- Developed practical experience in Python-based data analytics.
- Strengthened Pandas and NumPy skills for data preparation.
- Applied exploratory data analysis to a real-world style dataset.
- Designed interactive Plotly visualizations.
- Built a multi-page Streamlit dashboard.
- Practiced feature engineering, normalization, aggregation and trend analysis.
- Improved business-oriented data storytelling.
- Gained experience debugging data-loading, visualization and dashboard integration issues.
- Learned the importance of documentation, reproducibility and analytical limitations.

14.2 Future Enhancements

- Integrate a larger and more current application dataset.
- Add application-level search and drill-through analysis.
- Introduce robust time-series forecasting with validation.
- Add machine-learning models for popularity or review-volume prediction.
- Integrate sentiment analysis from user reviews.
- Add automated data-quality monitoring.
- Deploy the dashboard to a cloud environment with controlled access.
- Add downloadable filtered reports and datasets.
- Improve performance through caching and pre-aggregated analytical tables.

14.3 Conclusion

The Google Play Store Analytics Dashboard demonstrates a complete data analytics workflow from raw data preparation to interactive business intelligence. By combining data cleaning, exploratory analysis, feature engineering, advanced visualization and Streamlit-based interaction, the project transforms complex application records into an accessible analytical environment.

The six visualization modules provide complementary perspectives: density analysis, hierarchical segmentation, calendar-based trend analysis, category flow analysis, normalized performance comparison and free-versus-paid profiling. Together, they demonstrate how visualization can support structured exploration and business-oriented interpretation.

The project also highlights an important principle of analytics: reliable conclusions depend on data quality, transparent assumptions, appropriate aggregation and careful interpretation. As an internship and portfolio project, the dashboard provides a practical demonstration of skills relevant to entry-level data analytics and data science work.

15. References

- Google Play Store application dataset used for the internship project.
- Python Documentation — Python programming language reference.
- Pandas Documentation — data structures and analytical operations.
- NumPy Documentation — numerical computing and array operations.
- Plotly Documentation — interactive data visualization framework.
- Streamlit Documentation — framework for building data applications.
- Microsoft Power BI Documentation — business intelligence concepts, where applicable.

End of Report